

Statistical Inference on Global Health Data: A comprehensive analysis of the WHO Life Expectancy Dataset

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The dataset was loaded successfully. We proceeded to do some checks onto the dataset.

```
# Checking for dimensions
cat("Dimensions:", dim(df), "\n\n")
```

```
## Dimensions: 2864 21
```

```
# Checking column names
names(df)
```

```
## [1] "Country"           "Region"
## [3] "Year"              "Infant_deaths"
## [5] "Under_five_deaths" "Adult_mortality"
## [7] "Alcohol_consumption" "Hepatitis_B"
## [9] "Measles"           "BMI"
## [11] "Polio"             "Diphtheria"
## [13] "Incidents_HIV"     "GDP_per_capita"
## [15] "Population_mln"    "Thinness_ten_nineteen_years"
## [17] "Thinness_five_nine_years" "Schooling"
## [19] "Economy_status_Developed" "Economy_status_Developing"
## [21] "Life_expectancy"
```

```
# Checking data types
str(df)
```

```
## 'data.frame':    2864 obs. of  21 variables:
## $ Country          : chr  "Turkiye" "Spain" "India" "Guyana" ...
## $ Region           : chr  "Middle East" "European Union" "Asia" "South America" ...
## $ Year              : int   2015 2015 2007 2006 2012 2006 2015 2000 2001 2008 ...
## $ Infant_deaths     : num   11.1 2.7 51.5 32.8 3.4 9.8 6.6 8.7 22 15.3 ...
## $ Under_five_deaths : num   13 3.3 67.9 40.5 4.3 11.2 8.2 10.1 26.1 17.8 ...
## $ Adult_mortality   : num  105.8 57.9 201.1 222.2 58 ...
## $ Alcohol_consumption : num   1.32 10.35 1.57 5.68 2.89 ...
## $ Hepatitis_B       : int   97 97 60 93 97 88 97 88 97 97 ...
## $ Measles           : int   65 94 35 74 89 86 97 99 87 92 ...
## $ BMI               : num   27.8 26 21.2 25.3 27 26.4 26.2 25.9 27.9 26.5 ...
## $ Polio             : int   97 97 67 92 94 89 97 99 97 96 ...
## $ Diphtheria        : int   97 97 64 93 94 89 97 99 99 90 ...
## $ Incidents_HIV     : num   0.08 0.09 0.13 0.79 0.08 0.16 0.08 0.08 0.13 0.43 ...
## $ GDP_per_capita    : int  11006 25742 1076 4146 33995 9110 9313 8971 3708 2235 ...
## $ Population_mln    : num   78.53 46.44 1183.21 0.75 7.91 ...
## $ Thinness_ten_nineteen_years: num   4.9 0.6 27.1 5.7 1.2 2 2.3 2.3 4 2.9 ...
## $ Thinness_five_nine_years : num   4.8 0.5 28 5.5 1.1 1.9 2.3 2.3 3.9 3.1 ...
## $ Schooling         : num   7.8 9.7 5 7.9 12.8 7.9 12 10.2 9.6 10.9 ...
## $ Economy_status_Developed : int   0 1 0 0 1 0 0 1 0 0 ...
## $ Economy_status_Developing : int   1 0 1 1 0 1 1 0 1 1 ...
## $ Life_expectancy   : num   76.5 82.8 65.4 67 81.7 78.2 71.2 71.2 71.9 68.7 ...
```

```
# Checking for null values
colSums(is.na(df))
```

```
##           Country           Region
##           0             0
##           Year           Infant_deaths
##           0             0
##           Under_five_deaths       Adult_mortality
##           0             0
##           Alcohol_consumption       Hepatitis_B
```

```
##           0           0
##           Measles           BMI
##           0           0
##           Polio           Diphtheria
##           0           0
##           Incidents_HIV           GDP_per_capita
##           0           0
##           Population_mln Thinness_ten_nineteen_years
##           0           0
##           Thinness_five_nine_years           Schooling
##           0           0
##           Economy_status_Developed Economy_status_Developing
##           0           0
##           Life_expectancy
##           0
```

We saw that there 2864 rows and 21 columns in the dataset.

There were no missing values in all the variables. In order to be sure, we also proceeded to check if there were rows labeled 'null'.

```
# Checking if there were values masked as 'null', 'NULL' or " "
cat("Searching for text based masked nulls \n")
```

```
## Searching for text based masked nulls
```

```
colSums(df == "null" | df == "NULL" | df == "", na.rm = TRUE)
```

```
##           Country           Region
##           0           0
##           Year           Infant_deaths
##           0           0
##           Under_five_deaths           Adult_mortality
##           0           0
##           Alcohol_consumption           Hepatitis_B
##           0           0
##           Measles           BMI
##           0           0
##           Polio           Diphtheria
##           0           0
##           Incidents_HIV           GDP_per_capita
##           0           0
##           Population_mln Thinness_ten_nineteen_years
##           0           0
##           Thinness_five_nine_years           Schooling
##           0           0
##           Economy_status_Developed Economy_status_Developing
##           0           0
##           Life_expectancy
##           0
```

```
# Checking for suspicious zeros (0) in the health metrics
cat("\n Investigating null values masquerading as 0s \n")
```

```
##
## Investigating null values masquerading as 0s
```

```
cat("Zeros in Life Expectancy:", sum(df$Life_expectancy == 0, na.rm = TRUE), "\n" )
```

```
## Zeros in Life Expectancy: 0
```

```

cat("Zeros in under 5 deaths:", sum(df$Under_five_deaths == 0, na.rm = TRUE), "\n")

## Zeros in under 5 deaths: 0

cat("Zeros in under Adult Mortality:", sum(df$Adult_mortality == 0, na.rm = TRUE), "\n")

## Zeros in under Adult Mortality: 0

cat("Zeros in under alcohol consumption:", sum(df$Alcohol_consumption == 0, na.rm = TRUE),
    "\n")

## Zeros in under alcohol consumption: 38

cat("Zeros in under BMI:", sum(df$BMI == 0, na.rm = TRUE), "\n")

## Zeros in under BMI: 0

cat("Zeros in under Life Expectancy:", sum(df$Life_expectancy == 0, na.rm = TRUE), "\n")

## Zeros in under Life Expectancy: 0

cat("Zeros in under GDP per capita:", sum(df$GDP_per_capita == 0, na.rm = TRUE), "\n")

## Zeros in under GDP per capita: 0

cat("Zeros in incidents in HIV:", sum(df$Incidents_HIV == 0, na.rm = TRUE), "\n")

## Zeros in incidents in HIV: 0

```

It was seen that there were no masked values as “null”, ‘null’ or an empty string.

Upon investigating the zeros, there was no suspicious activity of zeros. The selected columns for this check were guided by real world scenarios. This is such as no country could have a GDP of zero or an entire population could have a BMI of zero. We did not check for the various diseases because it is possible for the various developed countries to have incidences where there are no certain diseases because of their well developed health care system which can also be backed by their robust immunization procedures. With regards with alcohol where we obtained 38 zeros, we needed to identify if these countries came from the Muslim majority countries or countries under an Islamic regime of which according to their religious views, consumption of alcohol is prohibited.

```

# Checking which countries have an alcohol consumption of zero
unique(df$Country[df$Alcohol_consumption == 0])

```

```

## [1] "Saudi Arabia" "Kuwait"          "Somalia"          "Bangladesh"      "Mauritania"
## [6] "Afghanistan"  "Libya"

```

The Muslim majority hypothesis proved to be correct as the countries with 0 alcohol consumption came from Islamic nations.

We needed to check if the highly-developed countries indeed had no vaccine preventable diseases. We also proceeded to investigate if presence of zeros meant that no immunization coverage was done or it meant that there were no such cases.

```

# Checking if it was immunization coverage or case count
cat("Summary of disease columns \n")

```

```

## Summary of disease columns

summary(df[, c("Measles", "Polio", "Diphtheria")])

```

```

##      Measles      Polio      Diphtheria
##  Min.   :10.00  Min.   : 8.0  Min.    :16.00
##  1st Qu.:64.00  1st Qu.:81.0  1st Qu.:81.00

```

```
## Median :83.00 Median :93.0 Median :93.00
## Mean :77.34 Mean :86.5 Mean :86.27
## 3rd Qu.:93.00 3rd Qu.:97.0 3rd Qu.:97.00
## Max. :99.00 Max. :99.0 Max. :99.00
```

An inspection of these diseases proved that there was a strict maximum threshold of 99. For actual cases, this value would scale to higher figures such as tens of thousands. This evidently proved that these values were actual percentage coverage rather than actual case counts.

We needed to do an investigation of these percentages. This was to check whether the percentages meant 99% covered or not covered. This was done by the use of a correlation. If it meant covered, there would be an inverse relationship between the disease and Infant_deaths or Under_five_deaths.

```
cat("The correlation between the Measles disease (coverage) and death of
  children under 5 years is", cor(df$Measles, df$Under_five_deaths, use = 'complete.obs'),
  "\n\n")
```

```
## The correlation between the Measles disease (coverage) and death of
## children under 5 years is -0.5129717
```

```
cat("The correlation between the Polio disease (coverage) and death of
  infants is", cor(df$Polio, df$Infant_deaths, use = 'complete.obs'), "\n\n")
```

```
## The correlation between the Polio disease (coverage) and death of
## infants is -0.7407905
```

This showed an inverse relationship between the diseases and the child mortality. This meant that the values exhibited by the diseases meant the extensiveness of the vaccinations. A 99% coverage meant that most parts of this particular country was vaccinated against the specific disease.

The project's next step was to check if indeed the high immunization coverage came from the highly developed countries. This would validate our hypothesis of developed countries having better healthcare. We began by investigating the robustness of Economy_status_Developing and Economy_status_Developed features to ensure that they were mutually exclusive.

```
# Investigating that the two economy columns do not add up to 2. Mutual
# exclusivity is needed
economy_check <- df$Economy_status_Developing + df$Economy_status_Developed
```

```
# Summary of the calculation
cat("Summary of the economy checks \n")
```

```
## Summary of the economy checks
```

```
summary(economy_check)
```

```
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
```

```
# Tabular form of these two columns
cat("\n Cross validation table of the two economy features \n")
```

```
##
```

```
## Cross validation table of the two economy features
```

```
table(Developed = df$Economy_status_Developed, Developing = df$Economy_status_Developing)
```

```
## Developing
## Developed 0 1
```

```
##          0      0 2272
##          1   592      0
```

We saw that the two columns were indeed mutually exclusive as they both did not add up to 2. The cross validation table backs this up whereby there are no individual countries where they are both developed and developing. This is given by the 0. Additionally, the table also informs us that there are 2272 records from developing countries and there are 592 records from developed countries.

We now proceeded to see if the high vaccination coverage came from developed or developing countries.